

Licensing Telecommunications Services

- Introduction
- Licensing Objectives
- Scarce Resources



Introduction

- ❖ Regulators issue Licenses to companies to
 - to operate telecommunications networks.
 - to provide telecommunications services.
- ❖ Licenses specify what facilities (i.e. networks and equipment) they can deploy, what services they can offer customers, and how they must interact with other businesses within the market.



Introduction

- ❖ Licenses may also specify access to scarce resources such as spectrum and telephone numbers.
- ❖ Companies pay fees for licenses, which may include both upfront fees, when a license is awarded, and annual payments.
- ❖ Major licenses usually have termination dates, before the end of which they and their conditions should be reviewed.



Licensing

- ❖ Historically, licenses have been allocated for different sub-markets within the overall telecoms market.
- ❖ E.g., to operate a fixed telephone network, or to build and operate a mobile wireless network, or to provide particular services such as paging, payphones or the internet.



Licensing

- ❖ These licenses differentiate between different technologies and services.
- ❖ Due to recent technological convergence different technology platforms can now deliver a wider range of services.
- ❖ Some regulators have therefore moved towards technology- and service-neutral licensing.



Licensing

- ❖ Technology-neutral licenses, for example, may permit a company to provide a telephone service without specifying that it has to use a fixed network or to use a particular wireless technology to do so: that choice is left up to the company.
- ❖ However, spectrum needs to be allocated through agreements such as licenses, in order to ensure that it is properly distributed and that there is no interference between spectrum users.



Licensing

- ❖ Two main types of licenses are awarded to companies.
These are;
 - Individual operator licenses
 - Class licenses (or general authorizations)



Individual operator licenses

- ❖ Individual operator licenses are specific to one operator or company.
- ❖ They are used primarily where
 - the number of licenses in the particular market is limited
 - there is likely to be market dominance
 - specific resources need to be allocated (e.g., in the mobile cellular sector, where typically there are no more than five operators in a country, each making use of radio spectrum.)



Class licenses or general authorizations

- ❖ Class licenses or general authorizations set out identical terms of operation for all businesses operating in a particular market segment, for example, internet service providers (ISPs), without attaching those licenses to a specific company.
- ❖ The current trend is now in favour of replacing individual operator licenses with general authorizations wherever possible
- ❖ The reason is that, issuing class licenses simplifies regulation, helps to ensure a level playing field, and is more flexible in handling technological and market change.



Licensing Objectives

- ❖ Governments and regulators normally have several different objectives for licensing telecommunications operators.
- ❖ Common licensing objectives are set out below:
 - Regulating Provision of an Essential Public Service
 - Expansion of Networks and Services and Other Universal Service Objectives
 - Privatization or Commercialization
 - Regulating Market Structure



Licensing Objectives

- Establishing a Competition Framework
- Allocation of Scarce Resources
- Generating Government Revenues
- Consumer Protection
- Regulatory Certainty



Regulating Provision of an Essential Public Service

- ❖ Basic telecommunications is viewed as an essential public service in most countries.
- ❖ While there has been an irreversible trend toward privatization and reliance on market forces, most governments continue to impose some controls to ensure basic telecommunications services are provided in the public interest.
- ❖ Licenses are an important tool for exercising such control in most countries.



Expansion of Networks and Services and Other Universal Service Objectives

- ❖ This is a major reason for licensing new telecommunications operators in most countries.
 - Network roll-out and service coverage obligations are often included in licenses.
 - This is particularly the case where a state-owned incumbent operator is privatized, or some degree of exclusivity is granted (e.g. a duopoly cellular license, with a right to use scarce spectrum).
- ❖ Licenses are an important tool for expanding infrastructure investment and promoting universal service and universal access objectives in developing countries.



Privatization or Commercialization

- ❖ A license is necessary where a state-owned incumbent is privatized.
- ❖ The license specifies the rights and obligations of the operator.
- ❖ It is a key document in the privatization process.
- ❖ It specifies what the investor is buying and what the government expects from the operator and the investor.



Regulating Market Structure

- ❖ A key aspect of regulation is the determination of the market structure of the telecommunications sector
 - in particular, the number of operators licensed to provide telecommunications services.
- ❖ In many countries a prime reason for licensing new telecommunications operators is to increase competition.
- ❖ Licensing requirements can also provide a means to limit market access.



Establishing a Competition Framework

- ❖ Licenses frequently include conditions to establish a “level playing field” for competition, and to limit the prospects that incumbent operators will abuse their dominant position in telecommunications markets.
- ❖ Such conditions are generally referred to in licenses as “anti-competitive safeguards” or “fair trading conditions”.



Allocation of Scarce Resources

- ❖ Finite (scarce) resources required in the operation of a telecommunications service include
 - radio spectrum
 - numbers
 - rights of way
- ❖ Such resources should be allocated between operators fairly, efficiently and in the public interest.



Allocation of Scarce Resources

- ❖ This allocation often requires a balancing of competing interests and priorities.
 - Spectrum, for instance, may be auctioned to the highest bidder or allocated at low cost to reduce prices or to encourage the rollout of new services.
 - Access to rights of way can be a source of revenue to government authorities or public utilities, but economic or other restrictions on access can delay the rollout of services and lead to higher consumer prices.



Generating Government Revenues

- ❖ Licensing of telecommunications operators and of radio spectrum can provide significant revenues to governments.
- ❖ An auction for new licenses can generate one-time revenues.
- ❖ In addition, annual license fees often provide a continuing source of revenue to fund the operations of the regulator, or for other purposes.
- ❖ In addition, licensing of new operators can increase the overall size of telecommunications markets and thus generate higher tax revenues for governments.



Consumer Protection

- ❖ Conditions relating to consumer protection are often included in telecommunications licenses.
- ❖ Such conditions may relate to matters such as
 - price regulation
 - billing practices



Consumer Protection

- ❖ Consumer complaint mechanisms
- ❖ Dispute resolution
- ❖ Limitations of liability for service defaults
- ❖ Mandatory services to consumers
 - e.g. directory services, operator assistance and emergency services



Regulatory Certainty

- ❖ By clearly defining the rights and obligations of the operator and the regulator, a license can significantly increase confidence in the regulatory regime.
- ❖ Regulatory certainty is a critical element of the licensing processes where the aim is to attract new operators and investment.
- ❖ This is particularly true in the case when foreign investment is sought in riskier developing or transitional economies.



Scarce Resources

- ❖ There are two main scarce resources in telecommunications: spectrum and numbers.
- ❖ **Spectrum** is the radio frequency required to provide services through wireless technologies
- ❖ The allocation and management of spectrum go beyond the telecom regulator's own responsibilities.



Scarce Resources

- In many countries, the security services play an important role in dictating spectrum use; or there may be a separate spectrum regulatory authority.
- The International Telecommunication Union (ITU) also manages the rules governing the international allocation of spectrum, which are intended to maximize its use while minimizing cross-border interference.



Scarce Resources

- ❖ **Numbers** are the other scarce resource in telecommunications.
- ❖ Although these numbers are infinite, they are considered scarce resource in telecommunications because only a limited number can readily be used, for two reasons.
 - People can only hold a limited number of digits ten or so in their short-term memories when dialing.



Scarce Resources

- More importantly, numbers have to be allocated in blocks which indicate what type of number they are (e.g., local numbers, international numbers, mobile cellular numbers, premium rate numbers) and what charges users can expect to incur by using them.
- ❖ There are international norms for numbering just as there are international rules for spectrum.



Scarce Resources

- ❖ Every country needs a numbering plan that is capable of managing increasing demand for numbers over a significant period of time as the number of customers increases and/or more services requiring numbers are offered.
- ❖ Numbering raises two main issues from the point of view of competition regulation.



Scarce Resources

- numbers need to be available to all competitors on equal terms, just like spectrum.
- leaving the ownership of numbers in the hands of operators inhibits competition.
- ❖ If a subscriber e.g., a small business has to change its telephone number when it changes service provider, it can incur significant transaction costs.



Scarce Resources

- ❖ Regulators can deal with this by introducing number portability, i.e. making the subscriber rather than the operating company the owner of a number, so that the subscriber can port (take) the number from one service provider to another.
- ❖ Number portability is increasingly regarded by regulators as an important mechanism for making telecommunications markets work effectively for the consumer.



Assignment (GROUP)

- Explain 3 methods used in spectrum licensing.
- What are the effects of such methods?
- Explain some methods used to enhance spectral efficiency.



Interconnection

- ❖ The core principle of interconnection is that everyone should be able to communicate with everyone else, irrespective of which network they subscribe to.

